Harvard

Fundamentals of Neuroscience · MCB80x

Formula & Constants Reference

Quick reference for proctored-exam calculations · 37 °C unless noted

Symbols follow course usage: V_m = membrane potential; E_{ion} = reversal (Nernst) potential for that ion; g = conductance (S), R = resistance (Ω), C_m = membrane capacitance. Outward current is often defined positive.

Constants (approximate)

Quantity	Value	Use
RT/F at 37 °C	≈ 26.7 mV	Nernst with natural log (ln)
2.303·RT/F at 37 °C	≈ 61.5 mV per 10× ratio	Nernst with $\log_{10}$
Typical $[K^+]_{in}$ / $[K^+]_{out}$	≈ 140 / 4–5 mM	E_K strongly negative
Typical $[Na^+]_{out}$ / $[Na^+]_{in}$	≈ 145 / 15 mM	E_{Na} strongly positive

Part 1 — Electrical properties

Nernst (equilibrium) potential — monovalent cation

$E_{ion} = (RT / zF) \cdot \ln([ion]_{out} / [ion]_{in})$

Log_{10} form: $E_{ion} = (61.5\ mV / z) \cdot \log_{10}([ion]_{out} / [ion]_{in})$ at 37 °C

Anions (e.g. Cl^-, z = −1): use the same magnitude with inside/outside in the log; sign follows your convention for E_{Cl}. Divalent Ca^{2+}: divide the 61.5 mV slope by 2. Doubling [out] shifts E by ≈ +18 mV (z = 1) or ≈ +9 mV per 2× for z = 2.

Goldman (constant-field) resting potential

$V_m = (RT/F) \cdot \ln((P_K[K^+]_{out} + P_{Na}[Na^+]_{out} + \ldots) / (P_K[K^+]_{in} + P_{Na}[Na^+]_{in} + \ldots))$

With only K^+ and Na^+ and $P_{Cl} \approx 0$, compute E_K, E_{Na} first, then apply permeability weights. Resting V_m lies between E_K and E_{Na}, nearer the ion with higher P.

Ohm's law for an open ion channel

$I_{ion} = g_{ion} \cdot (V_m - E_{ion})$

Driving force = $(V_m - E_{ion})$. At $V_m = E_{ion}$, net ionic current through that conductance is zero. Example: 50 nS × (−70 mV) = −3.5 nA.

Membrane time constant

$\tau = R_m \cdot C_m$

Larger τ → slower voltage changes for a given current. If R_m doubles with fixed injected current I, steady-state $V = I \cdot R_m$ doubles and τ doubles.

Length (space) constant

$\lambda = \sqrt{(R_m / R_i)}$ (units: distance along cable)

R_m = specific membrane resistance × geometry; R_i = axial resistance. Higher leak (lower R_m) → shorter λ. Larger diameter → lower R_i → longer λ.

Membrane capacitance (parallel-plate model)

$C_m \propto A / d$ (area / membrane thickness)

Thicker membrane → lower capacitance per unit area. Myelin increases effective R_m and reduces capacitive load at internodes.

Steady-state passive response to step current

$V_\infty = I_{inject} \cdot R_{in}$ where R_{in} is input resistance

Part 1 — Sign checks (no calculator)

- Raising extracellular K^+ → E_K less negative → V_m depolarizes toward new E_K.
- Raising extracellular Na^+ → E_{Na} more positive.
- At rest, V_m negative: opening Na^+ channels → depolarization toward E_{Na}.
- Opening K^+ channels when $V_m > E_K$ → hyperpolarizing/outward K^+ current.
- Na^+/K^+ pump: 3 Na^+ out, 2 K^+ in per cycle → electrogenic, hyperpolarizing contribution.

Part 2 — Synapses & integration (useful relations)

Quantal (mini) postsynaptic events

Evoked PSP ≈ n · q (n vesicles released × mean mini amplitude q)

Stochastic n explains trial-to-trial variability. Facilitation/depression change effective n or release probability p.

Summation (conceptual)

$V_m(t)$ integrates EPSPs/IPSPs within τ; spatial sum at soma/AIS from multiple synapses

Inhibitory shunt: increased conductance lowers effective R_{in} and attenuates EPSPs (divisive + subtractive effects).

NMDA receptor (LTP gate)

Requires glutamate + postsynaptic depolarization to relieve Mg^{2+} block → Ca^{2+} entry

Part 3 — Sensory & misc.

- **Weber–Fechner (concept):** perceived intensity ∝ log(stimulus strength) — explains compressive coding.
- **Cochlea:** place code — basal = high f, apical = low f; not a single exam identity formula.
- **Retina:** photoreceptor dark current via cGMP-gated channels; light → ↓ cGMP → hyperpolarization.
- **Ohm for dB (auditory):** intensity level $IL = 10 \cdot \log_{10}(I/I_0)$ dB — know log-compression idea.

Unit reminders

- 1 S = 1 A/V; 1 nS × 1 mV = 1 pA; 1 nS × 70 mV = 70 pA = 0.07 nA.
- Capacitive current $I_C = C_m \cdot dV_m/dt$ (transient at step onset).

Personal MCB80x study reference — pair with the exam study guide and practice quizzes on this site. Not an official HarvardX/edX document. Printed: November 1, 2026